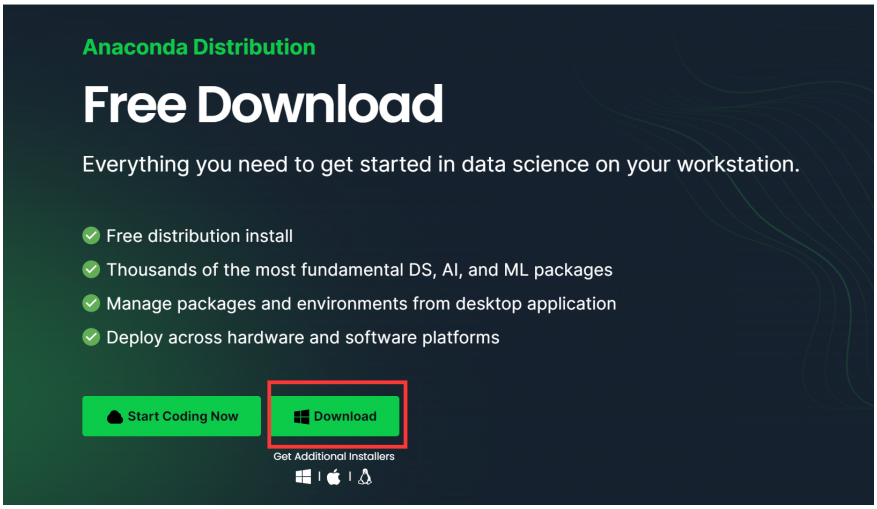
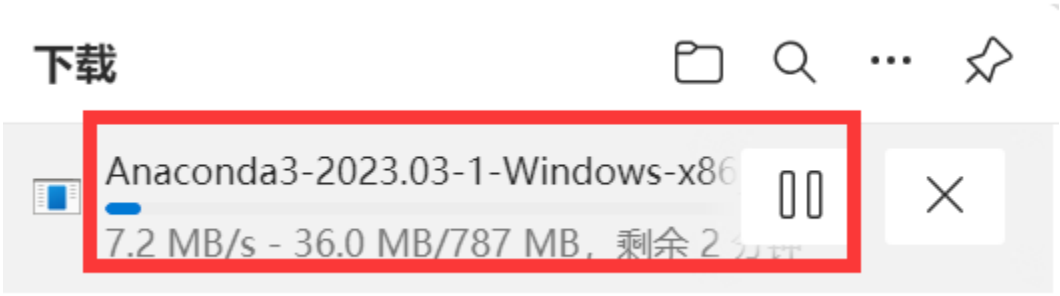
labelImg标注图片教程

在安装labelimg之前，我们需要先进行一个环境搭建，搭建环境之后再进行labelimg的安装和使用。

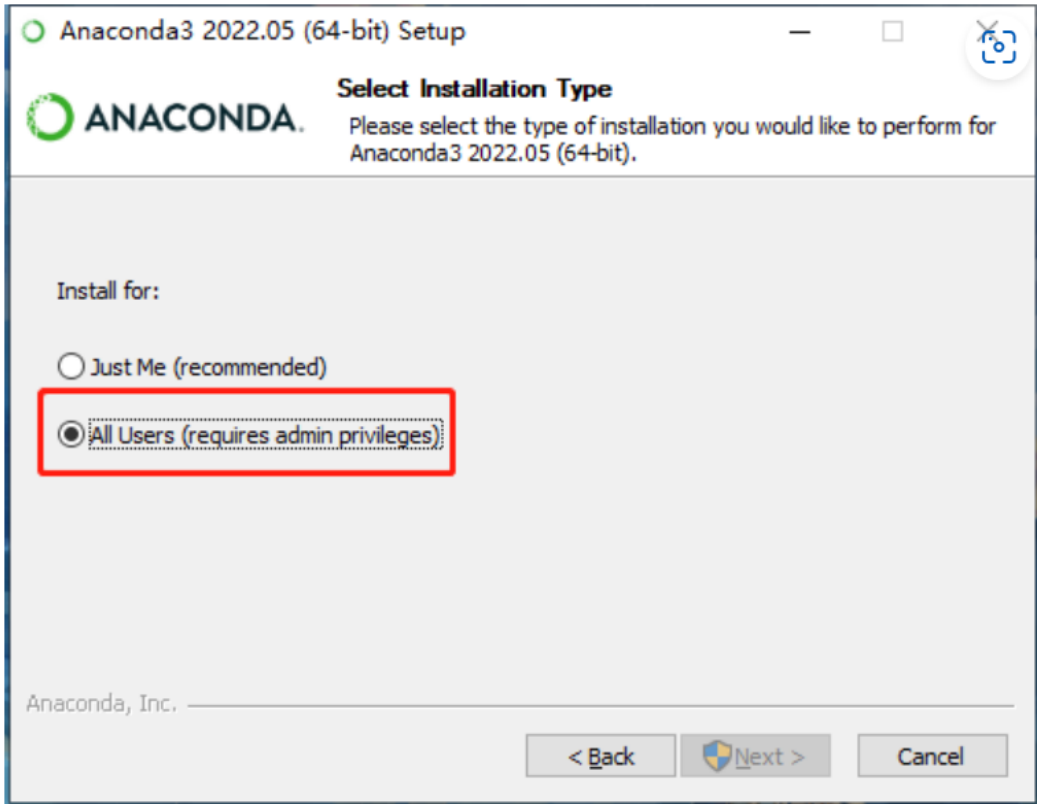
环境搭建：anaconda官网下载

官网下载地址链接

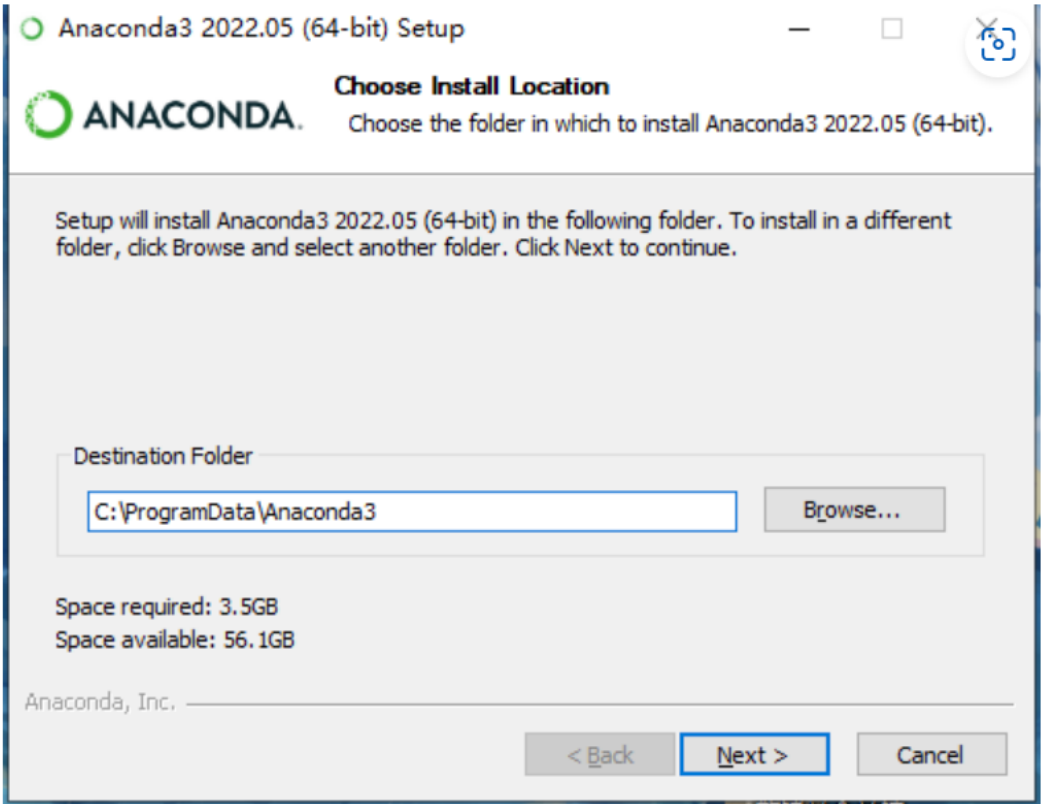




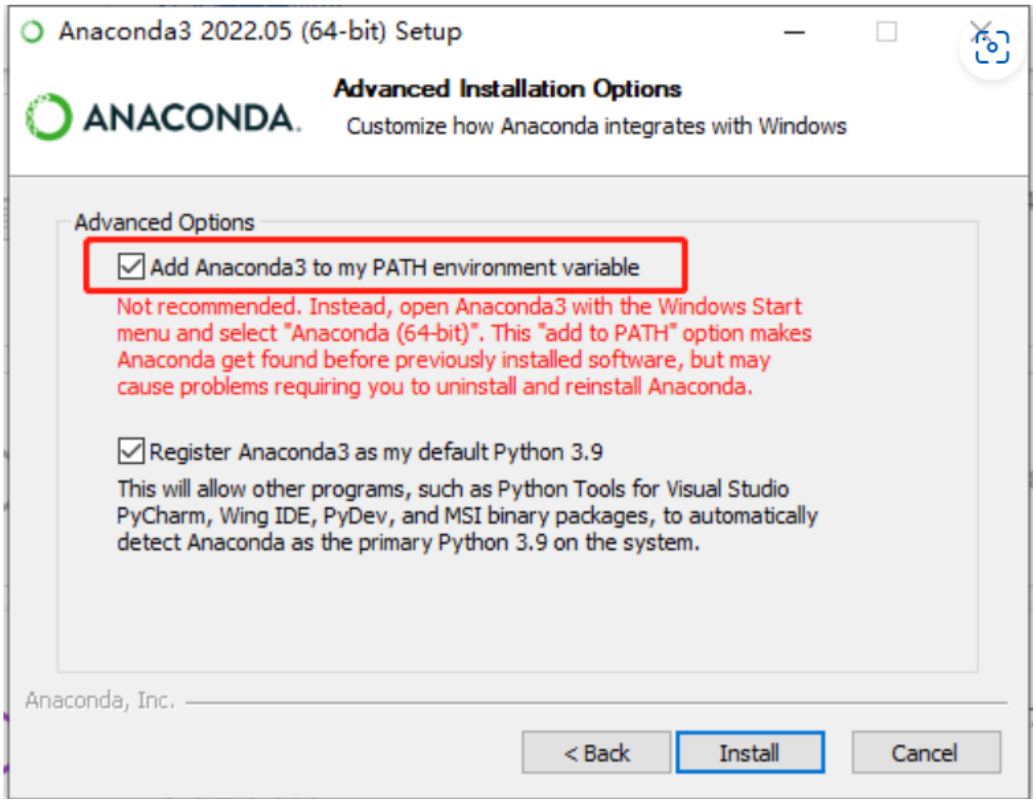
打开anaconda3的安装文件，进入安装界面，在install for安装选项中建议勾选all users，如下图。



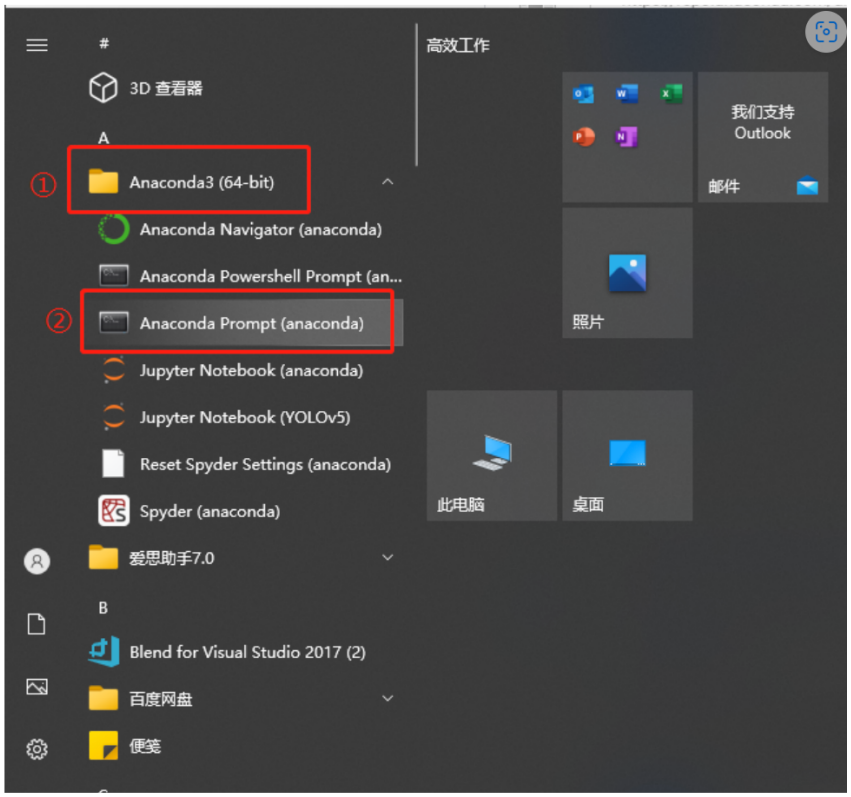
在设置安装路径时，不建议使用默认的安装路径，建议在其他的盘符中新建文件夹进行安装。



在advanced options中勾选第一个√，将anaconda3的安装路径加入环境变量中。



安装完成后，打开开始菜单栏，选择anaconda3-anaconda prompt进入命令行。



在命令行中输入下列代码，若如下图显示pip版本号则证明anaconda3安装成功。

pip --version

```
(python3.6) C:\Users\novo>pip --version
pip 21.2.2 from D:\anaconda\envs\python3.6\lib\site-packages\pip (python 3.6)
```

环境配置完成后就可以进行labelimg的安装啦！

安装labelimg

在命令行窗口中**依次输入**下列代码，安装labelimg依赖的第三方库。

pip install PyQt5

pip install pyqt5-tools

pip install lxml

pip install labelimg

（注：此处可能会因网络原因造成卡顿，请耐心等待，若因为网络问题安装失败则重试）

使用pip安装第三方库流程如下：

#安装PyQt5，安装成功截图如下

```
(python3.6) C:\Users\novo>pip install PyQt5
Collecting PyQt5
  Using cached PyQt5-5.15.6-cp36-abi3-win_amd64.whl (6.7 MB)
Collecting PyQt5-sip<13,>=12.8
  Downloading PyQt5_sip-12.9.1-cp36-cp36m-win_amd64.whl (83 kB)
     |████████████████████████████████| 83 kB 306 kB/s
Collecting PyQt5-Qt5>=5.15.2
  Downloading PyQt5_Qt5-5.15.2-py3-none-win_amd64.whl (50.1 MB)
     |████████████████████████████████| 50.1 MB 48 kB/s
Installing collected packages: PyQt5-sip, PyQt5-Qt5, PyQt5
Successfully installed PyQt5-5.15.6 PyQt5-Qt5-5.15.2 PyQt5-sip-12.9.1
```

#安装pyqt5-tools，安装成功截图如下

```
(python3.6) C:\Users\novo>pip install pyqt5-tools
Collecting pyqt5-tools
  Using cached pyqt5_tools-5.15.4.3.2-py3-none-any.whl (29 kB)
Collecting click
  Using cached click-8.0.4-py3-none-any.whl (97 kB)
Collecting python-dotenv
  Downloading python_dotenv-0.20.0-py3-none-any.whl (17 kB)
Collecting pyqt5-plugins<5.15.4.3,>=5.15.4.2.2
  Downloading pyqt5_plugins-5.15.4.2.2-cp36-cp36m-win_amd64.whl (67 kB)
     |████████████████████████████████| 67 kB 218 kB/s
Collecting pyqt5==5.15.4
  Using cached PyQt5-5.15.4-cp36.cp37.cp38.cp39-none-win_amd64.whl (6.8 MB)
Requirement already satisfied: PyQt5-sip<13,>=12.8 in d:\anaconda\envs\python3.6\lib\site-packages (from pyqt5==5.15.4->pyqt
5-tools) (12.9.1)
Requirement already satisfied: PyQt5-Qt5>=5.15 in d:\anaconda\envs\python3.6\lib\site-packages (from pyqt5==5.15.4->pyqt5-to
ols) (5.15.2)
Collecting qt5-tools<5.15.2.2,>=5.15.2.1.2
  Downloading qt5_tools-5.15.2.1.2-py3-none-any.whl (13 kB)
Collecting click
  Downloading click-7.1.2-py2.py3-none-any.whl (82 kB)
     |████████████████████████████████| 82 kB 48 kB/s
Collecting qt5-applications<5.15.2.3,>=5.15.2.2.2
  Downloading qt5_applications-5.15.2.2.2-py3-none-win_amd64.whl (60.9 MB)
     |████████████████████████████████| 60.9 MB 54 kB/s
Installing collected packages: qt5-applications, click, qt5-tools, pyqt5, python-dotenv, pyqt5-plugins, pyqt5-tools
  Attempting uninstall: pyqt5
    Found existing installation: PyQt5 5.15.6
    Uninstalling PyQt5-5.15.6:
      Successfully uninstalled PyQt5-5.15.6
Successfully installed click-7.1.2 pyqt5-5.15.4 pyqt5-plugins-5.15.4.2.2 pyqt5-tools-5.15.4.3.2 python-dotenv-0.20.0 qt5-app
lications-5.15.2.2.2 qt5-tools-5.15.2.1.2
```

#安装lxml，安装成功截图如下

```
(python3.6) C:\Users\novo>pip install lxml
Collecting lxml
  Downloading lxml-4.9.0-cp36-cp36m-win_amd64.whl (3.7 MB)
     |████████████████████████████████| 3.7 MB 29 kB/s
Installing collected packages: lxml
Successfully installed lxml-4.9.0
```

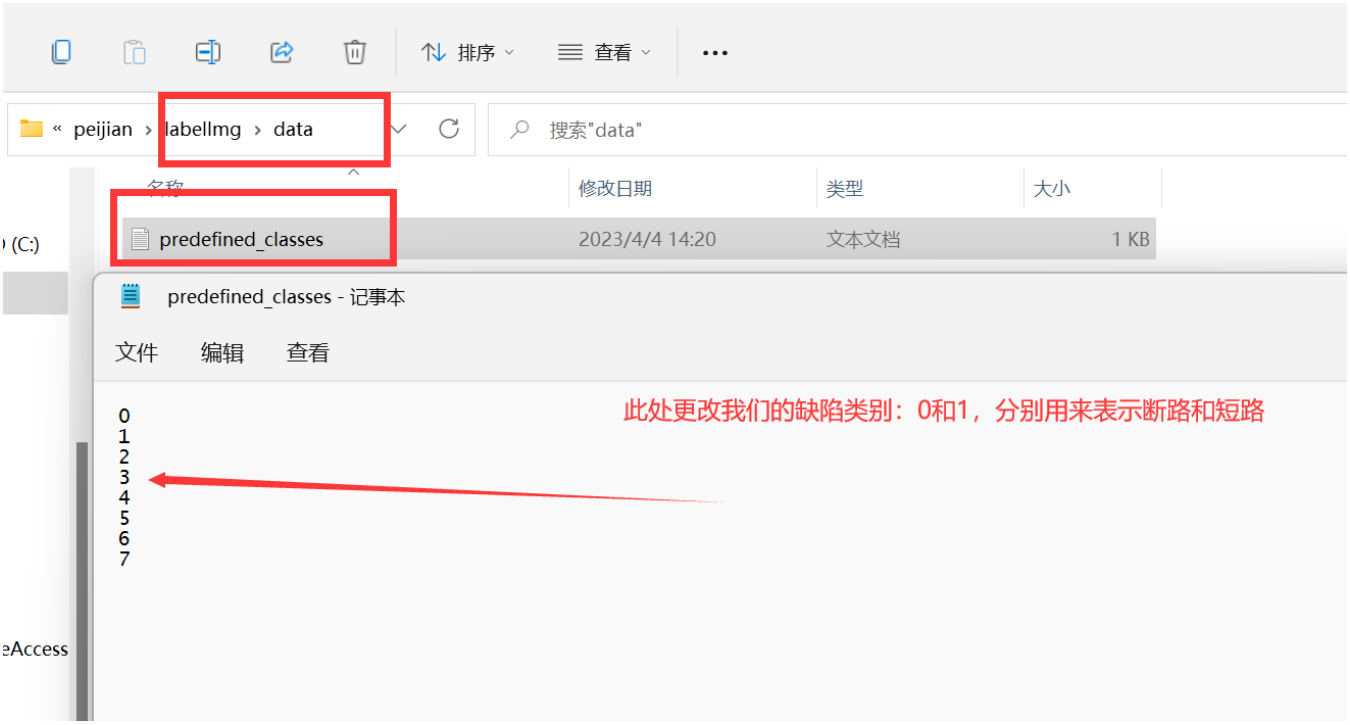
#安装labelimg，安装成功截图如下

```
(python3.6) C:\Users\novo>pip install labelimg
Collecting labelimg
  Using cached labelImg-1.8.6.tar.gz (247 kB)
Requirement already satisfied: pyqt5 in d:\anaconda\envs\python3.6\lib\site-packages (from labelimg) (5.15.4)
Requirement already satisfied: lxml in d:\anaconda\envs\python3.6\lib\site-packages (from labelimg) (4.9.0)
Requirement already satisfied: PyQt5-sip<13,>=12.8 in d:\anaconda\envs\python3.6\lib\site-packages (from pyqt5->labelimg) (1
2.9.1)
Requirement already satisfied: PyQt5-Qt5>=5.15 in d:\anaconda\envs\python3.6\lib\site-packages (from pyqt5->labelimg) (5.15.
2)
Building wheels for collected packages: labelimg
  Building wheel for labelimg (setup.py) ... done
  Created wheel for labelimg: filename=labelImg-1.8.6-py2.py3-none-any.whl size=261548 sha256=a562b106a56f7eed5ce0ccf7e26eae
0b949060fd0a823e27e9de2e175b2f214e
  Stored in directory: c:\users\novo\appdata\local\pip\cache\wheels\35\6f\d6\f96de77faaf1eab00bfe865ebe8806d3f790d609e4d7a15
f50
Successfully built labelimg
Installing collected packages: labelimg
Successfully installed labelimg-1.8.6
```

至此，环境配置步骤已经全部完成。

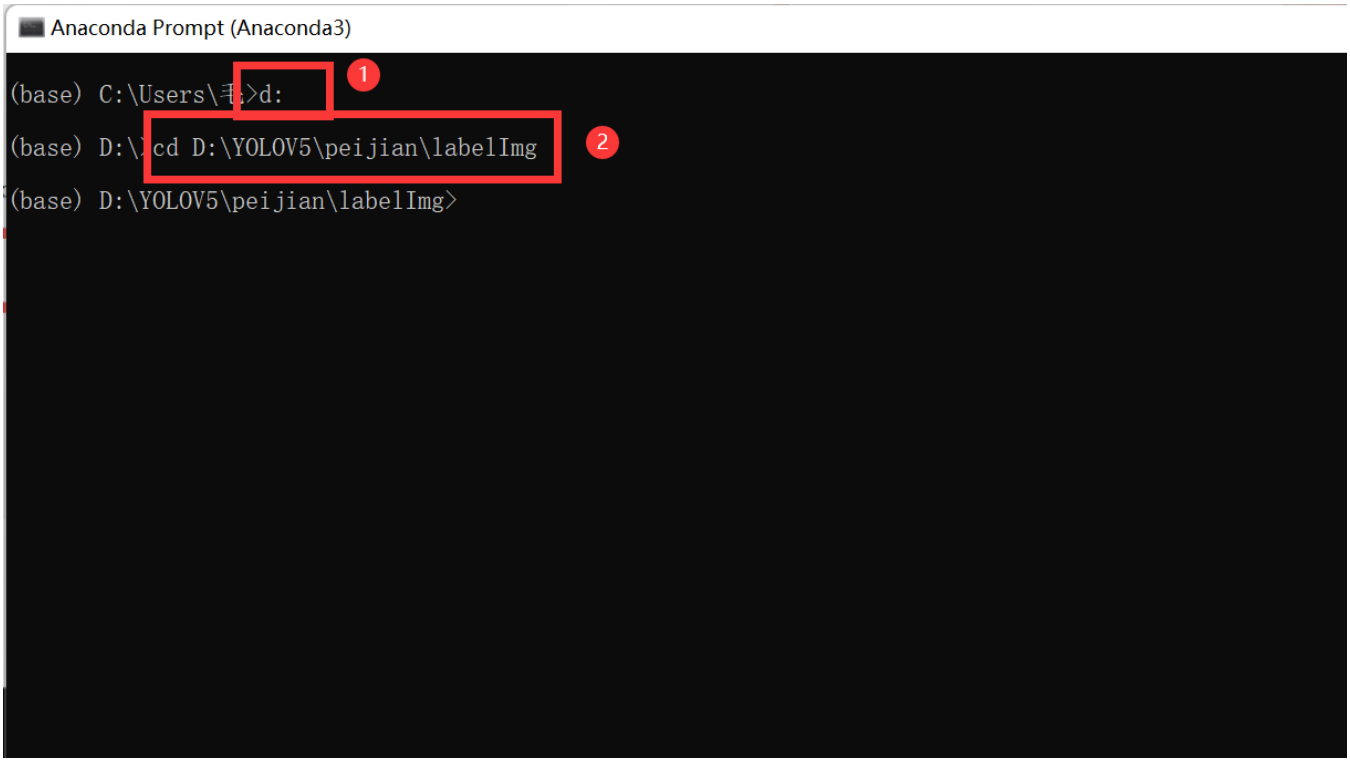
使用labellmg

运行软件前更改下要标注的类别。



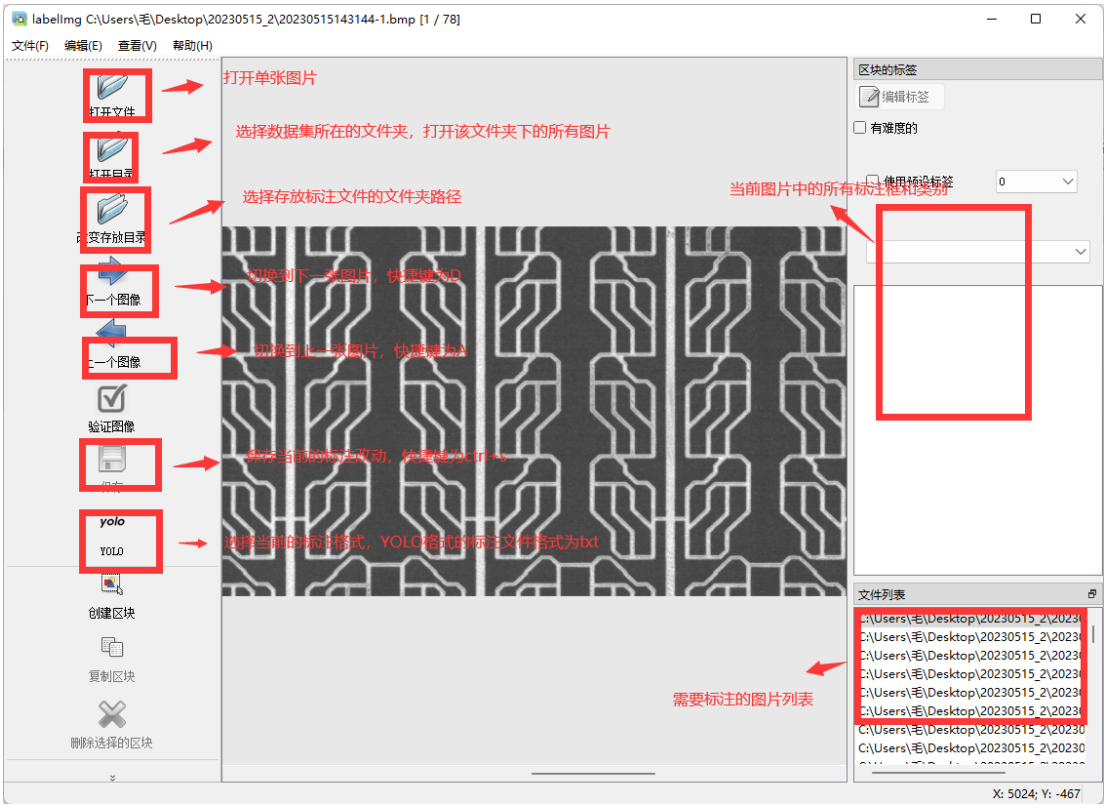
打开labellmg（要进入labellmg文件夹）

首先进入 labellmg文件夹

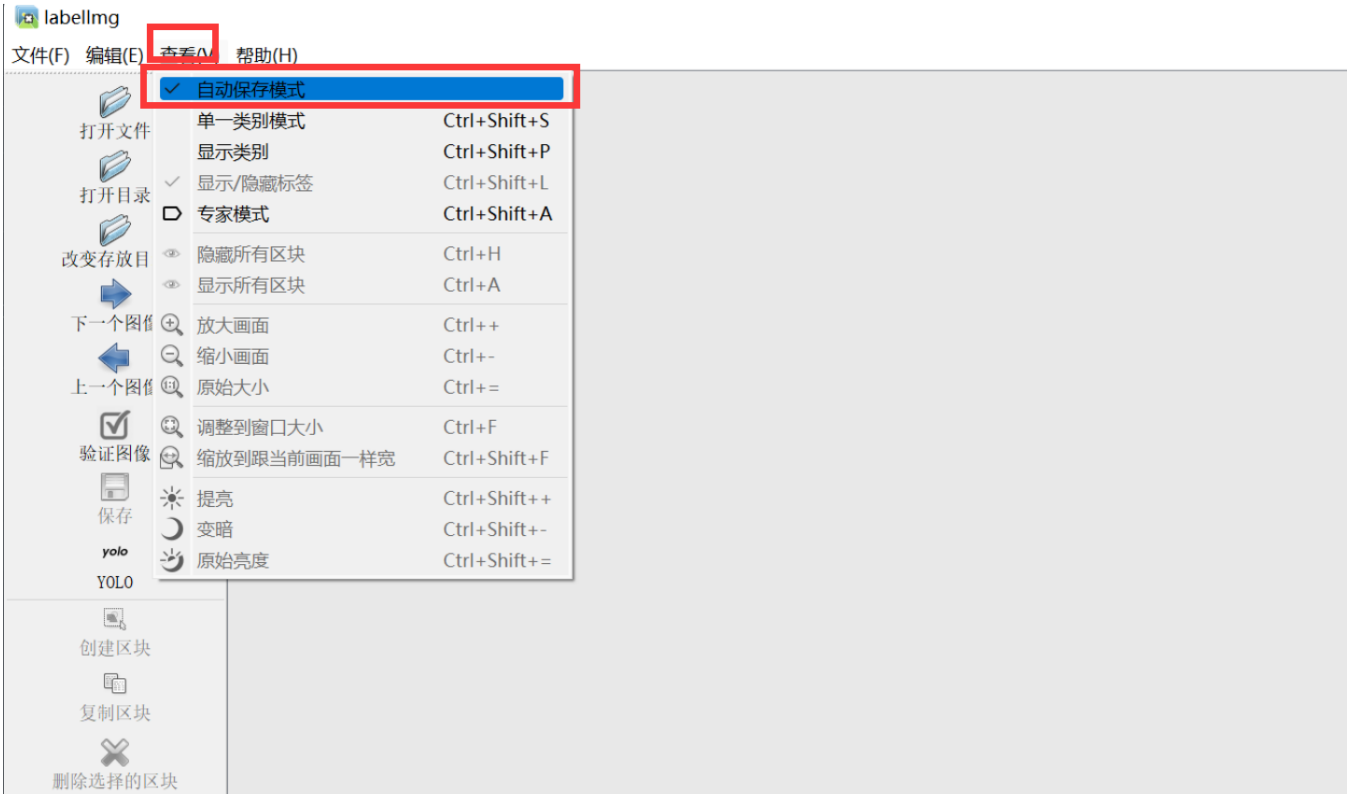


输入指令，运行软件：

python labelImg.py



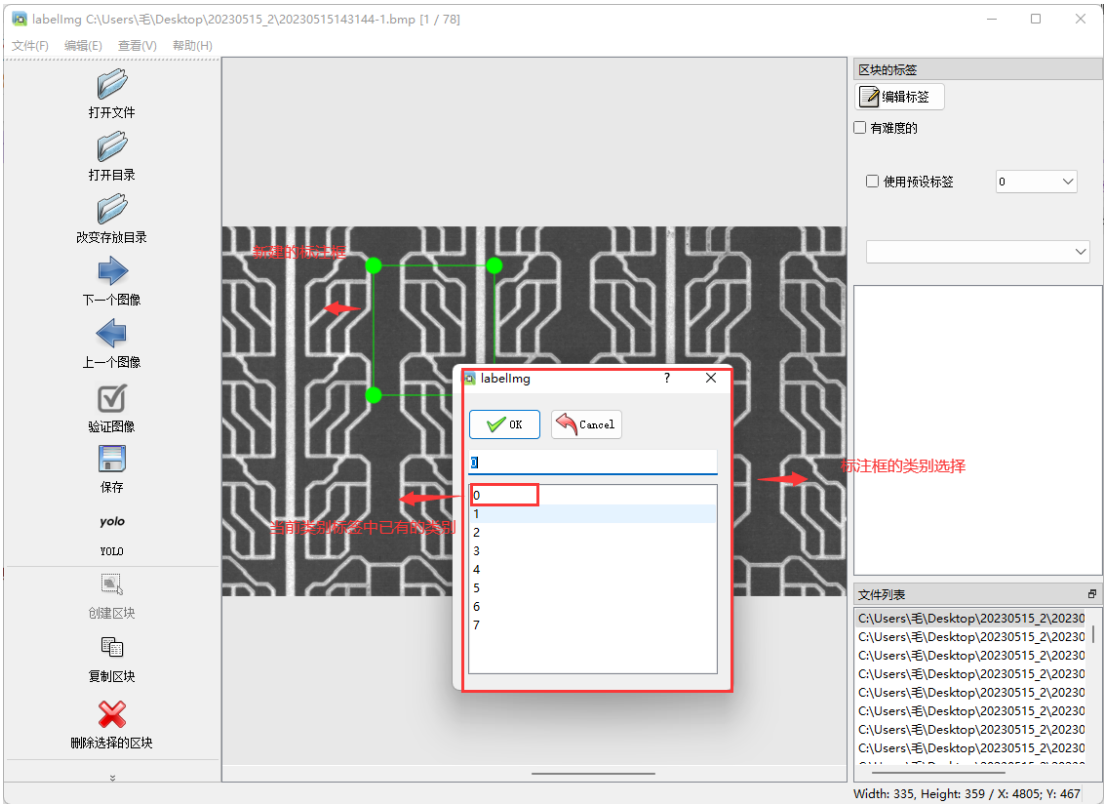
勾选 自动保存模式



标注技巧：

尽可能的完全拟合标注物体

在labelimg中，标注的快捷键为w，标注后用鼠标拖动锚框进行框选，框选完毕后会弹出类别选择框，如果是当前已有类则直接选择即可，若需要新加类别则在输入框中输入类别标签并确定。



(YOLO标注模式下)标注完成后数据集整体结构如下图所示。

